

Project Synopsis

1. Title

FemCare: Smart Pregnancy Care Companion

Team Details

Student Name	College	Branch	Mobile Number	Role
Gandham Sirisha	ISTS-Women's Engineering College- Rajahmundry	Computer Science and Engineering	9493011117	Team leader
Karimireddy Aswini Sowmya	ISTS-Women's Engineering College- Rajahmundry	Computer Science and Engineering	9100125498	Team Member
Ubalanka Kalavathi	ISTS Women's Engineering College- Rajahmundry	Computer Science and Engineering	9014992109	Team Member
Gandikota Jahnavi	ISTS Women's Engineering College	Computer Science and Engineering	6305152439	Team Member

Program Track and Domain

Select the appropriate Track and Domain for your project.

Program Track (Engg-HUB / Engg-Spoke / Science-HUB / Science-Spoke / Commerce-Spoke)	Domain (Analytics / Machine Learning / Deep Learning / GenAI)
Engg-HUB	Machine Learning

Guide Details

M Bharathi Raja

2. Problem Statement

Women often face difficulties in monitoring their health regularly, especially during pregnancy. Many women do not have continuous access to medical guidance, which can lead to health risks for both mother and baby. Traditional healthcare systems require frequent hospital visits, which may not always be possible due to distance, time, or lack of awareness.

Existing health applications provide general information but do not offer personalized tracking and monitoring features for women's health. There is a need for a system that can monitor health conditions, track fetal development, and provide useful guidance to pregnant women in a simple and accessible way.

To address this problem, the **FemCare system** is proposed. It is a digital healthcare platform designed to help women track pregnancy stages, monitor health conditions, and receive guidance regarding maternal care. The system aims to improve awareness, provide timely information, and support better healthcare management for women.

3. Objectives

The main objectives of the FemCare system are:

1. To develop a user-friendly web application for monitoring women's health.
2. To provide pregnancy tracking and fetal development information.
3. To allow users to record and manage their health data.
4. To provide health tips and guidance related to maternal care.
5. To develop a dashboard where users can monitor their health progress.
6. To improve awareness about pregnancy care and women's health.
7. To create a simple and accessible digital healthcare solution.

4. Methodology / Implementation Plan

The FemCare system is developed using a structured approach that includes system design, development, and testing.

First, the user registers in the system and creates a personal profile containing health details. The system securely stores this information in a database.

Once registered, users can log in and access the FemCare dashboard. The dashboard allows users to track pregnancy progress, monitor fetal development stages, and view important health tips.

The system provides pregnancy stage information and weekly fetal development updates to help women understand the growth of the baby. It also allows users to update their health information regularly.

The frontend of the system is developed using **HTML, CSS, and JavaScript**, which provides an interactive and user-friendly interface. The backend is developed using **Python**, which processes user data and manages system operations. A **MySQL database** is used to store and manage user information securely.

The system is tested to ensure accuracy, usability, and reliability.

5. Resources and Timeline

A. Hardware & Software Requirements

Hardware

- Laptop/PC
- Minimum 8 GB RAM
- Intel i5 Processor or above
- 256 GB SSD
- Stable Internet Connection

Software

- Operating System: Windows 10 / Windows 11
- Programming Languages: HTML, CSS, JavaScript, Python
- Tools: VS Code
- Database: MySQL

B. Dataset

The FemCare system does not require a large public dataset. Instead, the system stores user-entered health information.

Data Collection Method:

- Users enter their personal health information during registration.
- Pregnancy and fetal development information is collected from trusted medical resources.
- All data is stored securely for academic purposes.

C. Project Timeline:

Phase	Task	Tentative Completion Date
Phase 1	Problem Identification and Literature Survey	January 2026
Phase 2	System Design and Requirement Analysis	January 2026
Phase 3	System Development	February 2026
Phase 4	Testing and Debugging	March 2026
Phase 5	Documentation and Final Submission	March 2026

6. Expected Outcomes / Conclusion

The expected outcome of the FemCare project is a functional web-based healthcare application that helps women monitor their health and pregnancy stages. The system will provide useful information related to maternal care and fetal development.

The project aims to improve awareness about women's health and provide a simple digital solution for monitoring pregnancy. By using this system, women can easily track their health progress and receive guidance for better healthcare management.

7. References

1. Stuart Russell and Peter Norvig, **Artificial Intelligence: A Modern Approach**, Pearson.
2. Abraham Silberschatz, **Database System Concepts**, McGraw Hill.
3. Python Documentation – <https://www.python.org>
4. Women Health Organization resources on maternal health.
5. Medical resources on fetal development and pregnancy care.

